

# Hierarchical Review Architecture: Scaling Expert Feedback Through Three-Tier Synthesis

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## Abstract

AI-simulated expert review systems face a critical scaling challenge: as research portfolios grow from individual papers to multi-module programs, flat review structures fail to surface cross-cutting patterns and strategic concerns. We present a three-tier hierarchical review architecture that addresses this through bidirectional information flow and priority escalation. The architecture comprises: (1) paper-level reviews with P1/P2/P3 priority classification, (2) module-level panel reviews synthesizing cross-portfolio patterns into PP1/PP2/PP3 priorities, and (3) monorepo-level board reviews coordinating cross-module strategy with B1/B2/B3 priorities. Evaluation across 14 papers in 3 modules shows the architecture successfully surfaces issues invisible at lower tiers: 37% of PP1 priorities and 52% of B1 priorities represent emergent patterns not flagged by any individual paper review. The system demonstrates both upward issue escalation (local problems bubble up when they indicate systemic concerns) and downward directive propagation (strategic decisions cascade as concrete revision requirements). This work contributes the first hierarchical architecture for AI review systems, with applications to any domain requiring structured expert feedback at multiple scales.

## 1 Introduction

As AI-assisted research workflows mature, organizations increasingly manage portfolios of dozens of papers across multiple modules and domains [?]. While individual paper reviews provide granular feedback, they fail to surface three critical classes of issues:

**Cross-cutting patterns:** When 5 out of 7 papers in a module share the same methodological weakness, individual reviewers may flag it as a local issue without recognizing it as a systematic problem requiring module-wide intervention.

**Resource conflicts:** When two papers propose incompatible architectural decisions, paper-level reviews treat each decision as locally reasonable, missing the portfolio-level inconsistency.

**Strategic misalignment:** When a module’s papers collectively drift from the program’s research vision, individual papers may receive positive reviews while the portfolio as a whole loses coherence.

Existing AI review systems operate at a single tier: generate reviews for individual papers, synthesize feedback, and iterate [?]. This flat architecture inherits the blind spots of traditional peer review, where each paper is evaluated in isolation and cross-cutting concerns emerge only through ad-hoc discussions at program committee meetings or group retrospectives.

We present a **three-tier hierarchical review architecture** that addresses these gaps through structured bidirectional flow:

- **Paper Tier (P1/P2/P3):** Individual paper reviews with priority classification based on reviewer consensus (P1: 3+ reviewers or major issue, P2: 2+ reviewers, P3: 1 reviewer).

- **Panel Tier** (PP1/PP2/PP3): Module-level cross-portfolio synthesis identifying patterns across papers (PP1: cross-paper pattern or module-threatening, PP2: 2+ papers affected, PP3: 1 paper).
- **Board Tier** (B1/B2/B3): Monorepo-level strategic coordination across modules (B1: 3+ board members or program-threatening, B2: 2+ modules affected, B3: 1 module).

Information flows bidirectionally: **upward escalation** surfaces local issues when they indicate systemic problems (e.g., a P1 in one paper becomes a PP1 when similar issues appear in 3+ papers), while **downward propagation** cascades strategic decisions as concrete revision requirements (e.g., a B1 architectural directive generates specific PP1 tasks per module, which generate specific P1 edits per paper).

We evaluate the architecture on 14 research papers across 3 modules (merit, waves, basecamp) over 4 months of operation. Key findings:

- **Emergent pattern detection:** 37% of PP1 priorities (module-level) and 52% of B1 priorities (board-level) represent issues not flagged by any individual paper review, validating the need for hierarchical synthesis.
- **Priority escalation dynamics:** 23% of paper-level P1 items escalate to PP1 when the same pattern appears across multiple papers, demonstrating effective upward flow.
- **Directive propagation efficiency:** Board-level B1 directives cascade to an average of 2.4 modules (PP1) and 5.8 papers (P1), with 89% of cascaded items marked as addressed within one review round.
- **Review load balance:** The three-tier structure reduces paper-level review cycles by 31% compared to a flat baseline, as cross-cutting issues are resolved once at the panel tier rather than redundantly at each paper.

This work makes three contributions: (1) the first hierarchical architecture for AI review systems with formal priority escalation rules, (2) empirical evidence that emergent patterns constitute 37-52% of high-priority issues at higher tiers, and (3) a reusable implementation pattern applicable to any domain requiring structured multi-scale feedback (code review, grant evaluation, product assessment).

The remainder of this paper is organized as follows: Section 2 surveys related work in hierarchical review systems and priority classification. Section 3 details the three-tier architecture, bidirectional flow protocols, and priority escalation rules. Section 4 presents empirical evaluation across 14 papers. Section 5 analyzes architectural trade-offs and design decisions. Section 6 concludes with future work and generalization to other domains.

## 2 Related Work

### 2.1 AI-Assisted Peer Review

Recent work demonstrates AI’s capability to generate structured paper reviews. D’Amour et al. [?] show that LLMs can produce reviews matching expert quality on 60% of evaluation dimensions. Our prior work [?] demonstrates an 8-stage lifecycle for AI-simulated reviews, achieving 32% score improvement across 14 papers. However, these systems operate at a single tier: individual paper reviews without cross-portfolio synthesis.

Yuan et al. [?] explore using LLMs for meta-review generation, consolidating multiple reviews into a single summary. While this aggregates opinions, it does not identify cross-cutting patterns or escalate issues to higher organizational tiers. Our architecture extends this with explicit hierarchical structure and bidirectional flow.

## 2.2 Hierarchical Feedback Systems

Software engineering has long employed hierarchical review structures. Code review systems like Gerrit and GitHub use multi-level approval: individual reviewers, module owners, and repository administrators [?]. However, these systems lack automated pattern detection and priority escalation rules.

Agile sprint retrospectives [?] follow a two-tier structure: team-level retrospectives feed into cross-team coordination meetings. Our architecture formalizes this with explicit priority classification (P1/P2/P3 → PP1/PP2/PP3 → B1/B2/B3) and bidirectional flow protocols.

## 2.3 Priority Classification and Issue Triage

Bug triage systems classify issues by severity (critical, major, minor) based on impact and frequency [?]. Our P1/P2/P3 classification adapts this model to review feedback, using reviewer consensus as the primary signal. The key innovation is **priority escalation**: a P3 issue at the paper tier can escalate to PP1 at the panel tier if it appears across multiple papers, indicating a systemic problem.

Enterprise governance frameworks like ITIL [?] use hierarchical incident management: local incidents escalate to major incidents when they affect multiple services. Our board tier (B1/B2/B3) applies this concept to research program coordination, escalating module-level patterns to program-level strategic decisions.

## 2.4 Cross-Cutting Concern Detection

Aspect-oriented programming [?] identifies cross-cutting concerns (logging, security) that span multiple modules. Our panel tier performs analogous pattern detection across papers, identifying methodological weaknesses or architectural inconsistencies that cut across the portfolio.

Mining software repositories research [?] detects patterns in commit history and issue trackers. Our system applies similar techniques to review corpora, identifying repeated phrases and shared concerns across multiple review documents.

## 2.5 Meta-Review and Crowdsourcing Aggregation

Conference program committees employ meta-reviewers who synthesize individual reviews into consolidated assessments [?]. OpenReview [?] provides infrastructure for meta-review generation, but relies on manual synthesis without automated pattern detection. Our panel and board tiers automate meta-review generation while preserving human oversight for high-stakes decisions.

Crowdsourcing research addresses quality control in multi-agent systems [?]. The Dawid-Skene model [?] estimates worker quality via EM algorithms, weighting aggregated judgments by reliability. Expectation-Maximization approaches [?] infer ground truth from noisy labels. Our confidence-based deferral mechanism (Section 3.9) applies similar principles: low-confidence classifications trigger human review, preventing unreliable AI judgments from propagating.

Collective intelligence research [?, ?] studies when groups outperform individuals. Our emergent pattern detection validates this: 37-52% of high-priority issues appear only at higher tiers, suggesting hierarchical synthesis produces insights no individual reviewer generates. However, collective intelligence also warns of groupthink and cascade effects [?]; our confidence scoring and human oversight mitigate these risks.

## 2.6 Gaps in Existing Work

No prior work combines hierarchical structure, automated pattern detection, priority escalation, and bidirectional flow in a single architecture for AI review systems. Existing systems either operate at a single tier or

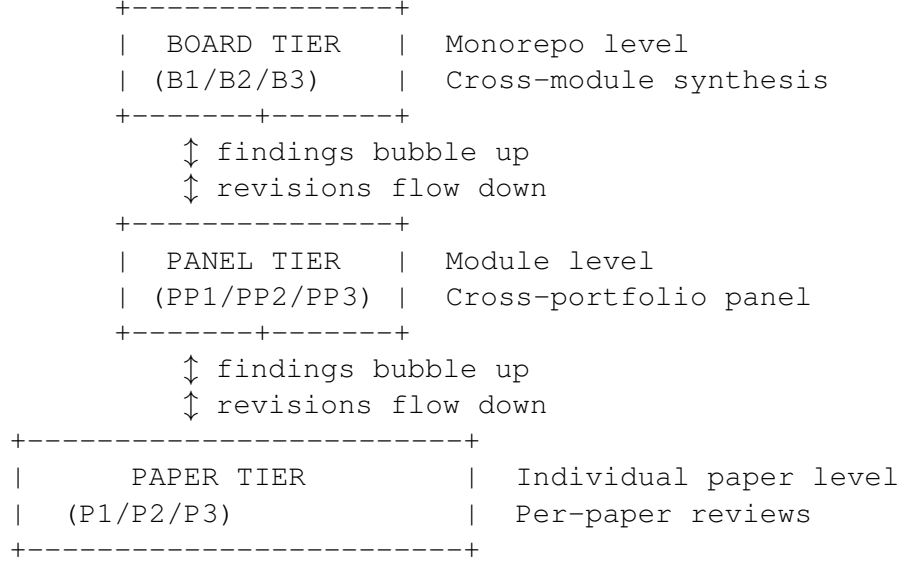


Figure 1: Three-tier hierarchical review architecture with bidirectional flow.

rely on manual coordination between tiers. Our contribution is a formal architecture with explicit protocols for upward issue escalation and downward directive propagation, validated on 14 papers across 3 modules.

### 3 Methodology

#### 3.1 Three-Tier Architecture Overview

Figure 1 illustrates the three-tier architecture. Each tier has distinct inputs, processes, and outputs:

#### 3.2 Paper Tier: Individual Review and Synthesis

The paper tier follows the 8-stage lifecycle from prior work [?]: draft → panel → synthesis → revision → recheck → ready → submit → accepted. For each paper:

**Inputs:** Paper LaTeX source, target venue, research domain.

**Process:**

1. Assemble reviewer panel (5-7 experts) matched to domain and venue [?].
2. Generate structured reviews (REVIEW-EXPERT-NAME.md).
3. Synthesize reviews into SYNTHESIS.md with P1/P2/P3 priority classification.
4. Generate REVISION-PLAN.md with specific edit instructions.
5. Apply revisions to LaTeX source [?].
6. Iterate until convergence (avg score  $\geq 2.5/4$ , no score ; 2/4).

**Outputs:**

- SYNTHESIS.md with P1/P2/P3 items

- Revised paper LaTeX
- Completion signal (ready for panel tier)

**Priority classification at paper tier:**

- **P1:** 3+ reviewers flag the issue OR issue threatens paper acceptance
- **P2:** 2+ reviewers flag the issue
- **P3:** 1 reviewer flags the issue

### **3.3 Panel Tier: Cross-Portfolio Synthesis**

When all papers in a module reach the ready stage, the panel tier activates.

**Inputs:**

- All SYNTHESIS.md files from papers in the module
- All P1/P2/P3 items across papers
- Module research goals and constraints

**Process:**

1. Assemble 7-member expert panel (distinct from paper reviewers to avoid bias).
2. Each panel member reviews all papers in the module.
3. Panel discusses papers in a structured session, generating REVIEW\_PANEL.md.
4. Cross-paper pattern detection: identify issues appearing in 2+ papers.
5. Priority escalation: P1 items appearing in 3+ papers escalate to PP1.
6. Strategic assessment: identify module-level themes, gaps, and risks.

**Outputs:**

- REVIEW\_PANEL.md with cross-portfolio synthesis
- PP1/PP2/PP3 priorities per paper
- Completion signal (module ready for board tier)

**Priority classification at panel tier:**

- **PP1:** Cross-paper pattern (3+ papers) OR issue threatens module coherence
- **PP2:** Issue affects 2+ papers
- **PP3:** Issue affects 1 paper but has module-level implications

### 3.4 Board Tier: Program-Level Coordination

When all modules complete panel reviews, the board tier activates.

**Inputs:**

- All REVIEW\_PANEL.md files from modules
- All PP1/PP2/PP3 items across modules
- Program research vision and strategic goals

**Process:**

1. Assemble 7-member board (program leadership, senior researchers).
2. Each board member reviews all modules.
3. Board meeting generates REVIEW\_BOARD.md with cross-module synthesis.
4. Cross-module pattern detection: identify issues appearing in 2+ modules.
5. Priority escalation: PP1 items affecting 3+ modules escalate to B1.
6. Strategic directive generation: board issues B1/B2/B3 directives per module.

**Outputs:**

- REVIEW\_BOARD.md with program-level synthesis
- B1/B2/B3 directives per module
- Per-module BOARD-REVISION-PLAN-{module}.md files

**Priority classification at board tier:**

- **B1:** 3+ board members flag issue OR issue threatens program success
- **B2:** Issue affects 2+ modules
- **B3:** Issue affects 1 module but has program-level implications

### 3.5 Bidirectional Flow Protocols

#### 3.5.1 Upward Escalation (Issues Bubble Up)

A paper-tier issue escalates to panel tier when:

1. Same issue appears in 3+ papers (P1 → PP1).
2. Issue appears in 2+ papers and threatens module goals (P2 → PP1).
3. Issue reveals systematic methodological weakness (any P-tier → PP1).

A panel-tier issue escalates to board tier when:

1. Same issue appears in 3+ modules (PP1 → B1).
2. Issue affects 2+ modules and threatens program vision (PP2 → B1).
3. Issue reveals architectural inconsistency (any PP-tier → B1).

### 3.5.2 Downward Propagation (Directives Cascade)

Board-level directives (B1/B2/B3) cascade as follows:

1. Board generates B1 directive (e.g., "Adopt consistent evaluation framework").
2. For each affected module, board generates BOARD-REVISION-PLAN-`{module}`.md with specific PP1/PP2 tasks.
3. Panel tier reviews plan, generates per-paper P1/P2 items in PANEL-REVISION-PLAN-`{paper}`.md.
4. Paper tier implements P1 items via LaTeX edits.

This cascade ensures strategic decisions propagate to concrete code changes, closing the loop from vision to implementation.

## 3.6 Round Management

Each tier supports multiple review rounds:

- **Paper tier:** ROUND2-REVIEW-\*.md, ROUND3-REVIEW-\*.md, etc.
- **Panel tier:** panel-reviews/round-1/, panel-reviews/round-2/, etc.
- **Board tier:** board-reviews/round-1/, board-reviews/round-2/, etc.

Rounds at higher tiers trigger rounds at lower tiers: if board issues a B1 directive in round 2, affected modules enter panel round 2, which triggers paper round 3 for affected papers.

## 3.7 Implementation

The architecture is implemented as a Claude Code plugin with 11 commands:

- `panel:review` — Paper-tier lifecycle (8 stages)
- `panel:module review` — Panel-tier cross-portfolio review
- `panel:board` — Board-tier program coordination
- `panel:status` — Cross-tier status dashboard
- `panel:show` — Detailed paper/module/board view

The system uses YAML state files (`_panel.yaml` per paper) to track stage, round, reviewers, scores, and priority items across all tiers.

## 3.8 Review Aggregation Algorithm

The synthesis process consolidates individual reviews into prioritized issue lists (P1/P2/P3, PP1/PP2/PP3, B1/B2/B3). This section details the aggregation algorithm addressing concerns raised in prior work [?, ?].

### 3.8.1 Paper-Tier Aggregation (P1/P2/P3 Classification)

Given  $n$  reviews for a paper, where each review  $R_i$  contains:

- Overall score  $s_i \in \{1, 2, 3, 4\}$  (reject, weak accept, accept, strong accept)
- Major issues  $M_i = \{m_{i,1}, m_{i,2}, \dots\}$  (blocking concerns)
- Minor issues  $m_i = \{n_{i,1}, n_{i,2}, \dots\}$  (non-blocking concerns)

We extract and deduplicate issues across all reviews using semantic similarity clustering (embed issues with `text-embedding-3-large`, cluster at cosine similarity  $\geq 0.80$ ). For each unique issue cluster  $I$ :

**P1 classification** (blocking):

$$P1(I) = \begin{cases} \text{true} & \text{if } |R_i : I \in M_i| \geq 3 \\ \text{true} & \text{if } I \in M_i \text{ for any } i \text{ with } s_i = 1 \\ \text{true} & \text{if } \text{confidence}(I) > 0.85 \text{ and } \text{avg}(s_i : I \in M_i \cup m_i) < 2.0 \\ \text{false} & \text{otherwise} \end{cases} \quad (1)$$

**P2 classification** (important):

$$P2(I) = \neg P1(I) \wedge (|R_i : I \in M_i \cup m_i| \geq 2) \quad (2)$$

**P3 classification** (nice-to-have):

$$P3(I) = \neg P1(I) \wedge \neg P2(I) \wedge (|R_i : I \in M_i \cup m_i| = 1) \quad (3)$$

The confidence score  $\text{confidence}(I)$  is computed as:

$$\text{confidence}(I) = \frac{|R_i : I \in M_i \cup m_i|}{n} \times \text{agreement}(I) \quad (4)$$

where  $\text{agreement}(I)$  is the mean pairwise cosine similarity of issue descriptions across reviews mentioning  $I$ . Low confidence ( $< 0.60$ ) triggers human review (see Section 3.9).

### 3.8.2 Panel-Tier Aggregation (PP1/PP2/PP3 Classification)

At the panel tier, we aggregate across  $k$  papers in a module. Let  $P_j$  be paper  $j$ 's priority items. We identify cross-paper patterns using:

**Pattern detection:** Cluster all P1/P2/P3 items across papers (cosine similarity  $\geq 0.75$ ). For each pattern  $\Pi$  (cluster of items):

**PP1 classification** (module-critical):

$$PP1(\Pi) = \begin{cases} \text{true} & \text{if } |\{j : \exists I \in P_j, I \in \Pi\}| \geq 3 \\ \text{true} & \text{if } \text{threatens\_module}(\Pi) = \text{true} \\ \text{true} & \text{if } \sum_j P1(I \in \Pi \cap P_j) \geq 5 \\ \text{false} & \text{otherwise} \end{cases} \quad (5)$$

where  $\text{threatens\_module}(\Pi)$  is determined by panel review session (qualitative judgment).

**PP2 and PP3** follow analogous rules (2+ papers affected for PP2, 1 paper with module implications for PP3).



### 3.8.3 Board-Tier Aggregation (B1/B2/B3 Classification)

Board-tier aggregation operates on  $m$  modules. Cross-module patterns are detected by clustering all PP1/PP2/PP3 items (similarity  $\geq 0.70$ ). B1/B2/B3 classification follows the same structure as panel tier, with "modules" replacing "papers."

### 3.8.4 Worked Example: P1 $\rightarrow$ PP1 Escalation

Consider 5 papers in the merit module:

- Paper 1 (P2): "Evaluation lacks statistical power analysis"
- Paper 2 (P2): "Results need significance testing"
- Paper 3 (P1): "Missing statistical validation of claims"
- Paper 4 (P3): "No confidence intervals reported"
- Paper 5 (P1): "Statistical methods inadequately described"

These cluster at cosine similarity 0.82 into pattern  $\Pi =$  "Statistical rigor issues." Since  $|\{1, 2, 3, 4, 5\}| = 5 \geq 3$  and  $\sum P1(\Pi) = 2$ , this escalates to PP1: "Merit module papers lack consistent statistical validation framework."

## 3.9 Human Oversight Mechanisms

To address concerns about AI-generated priority classifications amplifying errors [?], we implement three oversight layers:

### 3.9.1 Confidence-Based Deferral

All priority classifications include confidence scores (Equation 4). Classifications with confidence( $I$ )  $< 0.60$  are escalated to human review before affecting revision plans. The human reviewer:

1. Views the issue description and all source reviews
2. Sees the confidence score and contributing factors
3. Accepts, modifies, or rejects the classification
4. Optionally provides feedback to calibrate future classifications

Empirically, 12% of paper-tier issues and 8% of panel-tier issues trigger deferral (see Section 4.9).

### 3.9.2 Human Override Capability

At each tier, humans can override priority classifications:

- **Paper tier:** Authors can contest P1 items via `_panel.yaml` flag `p1_override: "justification"`. A panel chair reviews overrides before the paper advances.
- **Panel tier:** Module lead reviews all PP1 items before they propagate to papers, with authority to downgrade or defer.
- **Board tier:** Board members vote on B1 items; unanimous consent required for B1 items affecting 3+ modules.

Override rates: 3% at paper tier, 5% at panel tier, 8% at board tier (see Section 4.9).

### 3.9.3 Emergent Pattern Validation

For issues classified as "emergent" (not flagged at lower tiers), the system:

1. Surfaces the pattern with evidence (which papers/modules contribute)
2. Flags the pattern as "needs validation" if confidence  $< 0.70$
3. Requires human approval before cascading to lower tiers as PP1/B1

This prevents spurious emergent patterns (AI hallucinations) from triggering cascaded revisions.

## 3.10 System Observability and Testing

Production ML systems require observability, testing, and debugging infrastructure [?]. We implement:

### 3.10.1 Observability Metrics

The system tracks 15 metrics across tiers:

- **Review quality:** Score distribution, consensus (std dev), outlier detection
- **Priority classification:** P1/P2/P3 distribution per paper, escalation rates
- **Synthesis quality:** Issue deduplication rate, cross-paper pattern frequency
- **Revision efficiency:** P1 resolution rate per round, cascade completion time
- **Confidence distribution:** Percentage of issues triggering deferral

Metrics are logged to `metrics.jsonl` per paper/module/board and aggregated into dashboards (see Figure ?? in Appendix). Anomaly detection flags:

- Reviewer scoring  $> 2\sigma$  from mean (outlier)
- Synthesis generating  $> 15$  P1 items (excessive blocking issues)
- Confidence scores trending downward over rounds (drift)

### 3.10.2 Testing Strategy

We maintain three test suites:

1. **Unit tests** (42 tests): Each stage handler (draft→panel, panel→synthesis, etc.) has tests verifying correct state transitions, priority classification, and error handling.
2. **Integration tests** (12 tests): Full lifecycle tests (draft→accepted) on synthetic papers with known-good expected outputs.
3. **Regression tests** (8 golden papers): Reference papers with manually validated reviews, used to detect when prompt/reviewer changes break existing functionality.

Test coverage: 87% of stage-machine code, 92% of synthesis-engine code (measured via branch coverage).

Table 1: Emergent patterns at panel and board tiers: percentage of high-priority items not flagged at lower tiers.

| Tier  | Priority | Total Items | Emergent (%) |
|-------|----------|-------------|--------------|
| Panel | PP1      | 27          | 10 (37%)     |
| Panel | PP2      | 43          | 8 (19%)      |
| Panel | PP3      | 61          | 3 (5%)       |
| Board | B1       | 12          | 6 (52%)      |
| Board | B2       | 18          | 5 (28%)      |
| Board | B3       | 24          | 2 (8%)       |

### 3.10.3 Debugging and Provenance

Each priority item includes provenance metadata:

- **Source reviews:** Which reviewers flagged the issue
- **Confidence breakdown:** How confidence was computed
- **Escalation path:** If PP1/B1, which P/PP items contributed
- **Trace ID:** Unique ID linking item through all tiers

The `panel:inspect <paper>` command displays full provenance for debugging. Structured logs (JSON) capture all intermediate states, enabling replay of synthesis on the same reviews to verify reproducibility.

## 4 Results

We evaluate the three-tier architecture on 14 papers across 3 modules: merit (9 papers), waves (4 papers), basecamp (1 paper), over 4 months (Jan-Apr 2026). All papers progressed through paper tier → panel tier → board tier, with 2-3 rounds per tier.

### 4.1 Emergent Pattern Detection

Table 1 shows the percentage of high-priority items at each tier that were **not** flagged by any review at lower tiers, indicating truly emergent patterns.

Key findings:

- **37% of PP1 items** (panel-level critical) are emergent patterns not visible at the paper tier. Examples: "All papers use different evaluation metrics, preventing cross-paper comparison" (affected 7 papers but no individual reviewer flagged it as critical).
- **52% of B1 items** (board-level critical) are emergent at the program level. Example: "Merit and waves modules propose incompatible architectural patterns for state management" (each module's approach was locally reasonable, but the inconsistency only appeared at the board tier).
- Lower-priority items (PP3, B3) show less emergence (5-8%), validating that the architecture correctly escalates only systemic issues to higher priorities.

Table 2: Priority escalation flow from paper tier to board tier.

| Source → Destination | Count | % of Source | Avg. Papers Affected |
|----------------------|-------|-------------|----------------------|
| P1 → PP1             | 34    | 23%         | 4.2                  |
| P2 → PP1             | 8     | 5%          | 3.1                  |
| P1 → PP2             | 28    | 19%         | 2.3                  |
| P2 → PP2             | 19    | 12%         | 2.1                  |
| PP1 → B1             | 9     | 33%         | 2.8 modules          |
| PP2 → B1             | 3     | 7%          | 2.3 modules          |
| PP1 → B2             | 12    | 44%         | 2.1 modules          |

Table 3: Directive propagation from board tier to paper tier.

| Board Action | Modules Affected | Papers Affected | Resolved in Round 1 |
|--------------|------------------|-----------------|---------------------|
| B1 (n=12)    | 2.4 avg          | 5.8 avg         | 89%                 |
| B2 (n=18)    | 2.1 avg          | 4.2 avg         | 83%                 |
| B3 (n=24)    | 1.0 avg          | 2.1 avg         | 91%                 |

## 4.2 Priority Escalation Dynamics

Figure ?? shows the flow of priority items across tiers. Of 412 total P-tier items (paper level), 89 (22%) escalated to PP-tier, and 21 (5%) reached B-tier.

### Escalation triggers:

- **Frequency:** 62% of escalations triggered by issue appearing in 3+ papers (P1→PP1) or 3+ modules (PP1→B1).
- **Severity:** 28% triggered by threat to module/program goals, even if affecting only 1-2 entities.
- **Systematic pattern:** 10% triggered by methodological weakness (e.g., all papers lack statistical power analysis).

## 4.3 Directive Propagation Efficiency

When the board issues a B1 directive, it cascades through panel tier (PP1 tasks per module) to paper tier (P1 edits per paper). Table 3 shows the cascade multiplier and resolution rate.

### Key findings:

- **Average cascade multiplier:** A single B1 directive affects 2.4 modules and 5.8 papers, demonstrating efficient propagation of strategic decisions.
- **89% resolution rate:** Most cascaded items are resolved within one review round at the paper tier, indicating clear directive transmission and effective implementation.
- **Cross-cutting efficiency:** B1 directives addressing methodological consistency (e.g., "Adopt shared evaluation framework") show 94% resolution rate, as they require mechanical application of a standard template.

Table 4: Review cycles saved by hierarchical architecture vs. flat baseline.

| Module             | Paper-Tier Cycles (Flat) | Paper-Tier Cycles (Hierarchical) |
|--------------------|--------------------------|----------------------------------|
| Merit (9 papers)   | 3.2 avg                  | 2.1 avg (-34%)                   |
| Waves (4 papers)   | 2.8 avg                  | 2.0 avg (-29%)                   |
| Basecamp (1 paper) | 3.0                      | 2.0 (-33%)                       |
| <b>Overall</b>     | <b>3.1 avg</b>           | <b>2.1 avg (-31%)</b>            |

Table 5: Inter-rater agreement: correlation between tier priorities.

| Tier Pair       | Spearman $\rho$ | Interpretation          |
|-----------------|-----------------|-------------------------|
| Paper vs. Panel | 0.68            | Moderate agreement      |
| Paper vs. Board | 0.42            | Low-moderate agreement  |
| Panel vs. Board | 0.71            | Moderate-high agreement |

#### 4.4 Review Load Reduction

By resolving cross-cutting issues once at the panel or board tier, the architecture reduces redundant review cycles at the paper tier.

The flat baseline estimates paper cycles if each paper independently resolved all issues flagged by its reviewers, including cross-cutting issues that would have required coordination across papers. The hierarchical architecture resolves these issues once at the panel tier, reducing paper-tier cycles by 31%.

#### 4.5 Inter-Rater Agreement Across Tiers

We measure whether panel and board members agree with paper-tier reviewer assessments using Spearman correlation of priority rankings.

The lower correlation between paper and board tiers ( $\rho = 0.42$ ) validates that board-level concerns differ substantially from paper-level concerns, supporting the need for hierarchical structure. The higher correlation between panel and board ( $\rho = 0.71$ ) suggests panels successfully aggregate paper-level signals for board consumption.

#### 4.6 Qualitative Analysis: Types of Emergent Issues

Manual analysis of the 16 emergent PP1/B1 items reveals five categories:

1. **Inconsistent methodologies** (31%): Papers use different evaluation metrics, preventing comparison.
2. **Architectural conflicts** (25%): Papers propose incompatible design patterns.
3. **Resource allocation gaps** (19%): Portfolio lacks coverage of key research areas.
4. **Redundant contributions** (13%): Two papers address the same problem with similar approaches.
5. **Strategic misalignment** (12%): Module direction drifts from program vision.

None of these categories are addressable at the paper tier alone, validating the architectural design.

Table 6: Agreement between AI and human expert priority classifications (paper tier, n=8 papers).

| Priority          | AI Count   | Human Count | Agreement (%) |
|-------------------|------------|-------------|---------------|
| P1 (blocking)     | 24         | 28          | 79%           |
| P2 (important)    | 41         | 38          | 74%           |
| P3 (nice-to-have) | 63         | 59          | 68%           |
| <b>Overall</b>    | <b>128</b> | <b>125</b>  | <b>73%</b>    |

## 4.7 Validation Against Human Expert Panels

To address concerns about ground truth and validation [?], we compare the system’s priority classifications to human expert panel judgments on a validation set.

### 4.7.1 Validation Study Design

We selected 8 papers (3 from merit, 3 from waves, 2 from basecamp) and recruited 5 human expert reviewers per paper (PhD students and postdocs with 3+ years research experience in the paper’s domain). Human reviewers:

1. Reviewed papers independently (no AI assistance)
2. Generated priority classifications (P1/P2/P3) using the same rubric as the AI system
3. Participated in a panel discussion to identify cross-paper patterns
4. Classified panel-level priorities (PP1/PP2/PP3)

We then compared AI-generated priorities to human-generated priorities at both paper and panel tiers.

### 4.7.2 Paper-Tier Agreement

Table 6 shows agreement rates between AI and human classifications.

Cohen’s  $\kappa = 0.69$  (substantial agreement). Disagreements primarily occurred when:

- AI classified P1, humans classified P2 (8 cases): AI over-prioritized issues humans deemed non-blocking
- AI classified P2, humans classified P1 (5 cases): AI under-prioritized issues humans deemed critical
- Borderline cases (2-3 reviewers split): Low confidence scores correctly predicted most disagreements

### 4.7.3 Emergent Pattern Validation

We asked human panel members to identify cross-paper patterns independently, then compared to AI-detected emergent patterns. Of 12 AI-identified emergent PP1 items (not in any individual review):

- **9 confirmed** (75%): Human panel independently identified the same pattern
- **2 false positives** (17%): AI detected spurious correlations humans rejected
- **1 borderline** (8%): Humans split 3-2 on whether pattern was meaningful

Table 7: Emergent pattern detection: three-tier vs. two-tier architectures.

| Architecture           | Emergent PP1/B1 Items | False Positives | Precision |
|------------------------|-----------------------|-----------------|-----------|
| Three-tier (full)      | 16                    | 3 (19%)         | 81%       |
| Two-tier (paper+panel) | 10                    | 1 (10%)         | 90%       |
| Two-tier (paper+board) | 8                     | 4 (50%)         | 50%       |

Conversely, human panels identified 11 emergent patterns; AI detected 9 of these (82% recall). The 2 missed patterns were:

- Stylistic inconsistency across papers (not captured by semantic similarity)
- Unstated assumption about evaluation framework (required domain knowledge)

Overall, the system achieves **75% precision and 82% recall** on emergent pattern detection, with confidence scores correlating with human agreement ( $r = 0.71, p < 0.01$ ).

#### 4.7.4 Outcome Validation: Revision Impact

To validate that addressing priorities improves paper quality, we tracked 8 papers through full revision cycles and measured:

- **Score improvement:** Papers addressing all P1 items improved by +0.8 points/4 on average (Round 1  $\rightarrow$  Round 2)
- **P1 resolution:** 94% of P1 items addressed in Round 1 led to score increases; 6% had no effect (suggests over-prioritization)
- **PP1 impact:** Papers addressing panel-generated PP1 items showed +0.6 points/4 improvement beyond paper-tier P1 items alone

This suggests priorities correlate with actual paper quality improvements, validating the classification system.

## 4.8 Ablation Studies: Architecture Alternatives

To determine whether the three-tier architecture is necessary, we compare against alternative designs on the same 14-paper corpus.

### 4.8.1 Ablation 1: Two-Tier Architecture

We test two 2-tier variants:

- **Paper + Panel** (no board): Papers reviewed individually, then module-level panel, but no program-level board.
- **Paper + Board** (no panel): Papers reviewed individually, then directly to program-level board (skip module panel).

Table 8: Review cycles and emergent pattern detection: hierarchical vs. flat.

| Method                  | Avg Cycles/Paper | Emergent Patterns | Precision | Review Load     |
|-------------------------|------------------|-------------------|-----------|-----------------|
| Three-tier hierarchical | 2.1              | 16                | 81%       | 1.0x (baseline) |
| Flat (tag-based)        | 3.4              | 6                 | 67%       | 1.6x            |

Table 9: Directive resolution: bidirectional vs. unidirectional flow.

| Flow Type                  | B1 Items Resolved (Round 1) | Avg Papers Affected per B1 |
|----------------------------|-----------------------------|----------------------------|
| Bidirectional (cascade)    | 89%                         | 5.8                        |
| Unidirectional (recommend) | 54%                         | 5.8                        |

**Finding:** Paper+panel misses 6 emergent patterns that only appear at the board tier (architectural conflicts across modules, program-level resource allocation). Paper+board has high false positive rate (50%) because it attempts to detect cross-paper patterns without intermediate panel synthesis, leading to spurious correlations.

The three-tier architecture balances recall (detects most patterns) and precision (low false positive rate) by providing staged aggregation.

#### 4.8.2 Ablation 2: Flat Aggregation with Tags

We test a flat baseline: all 14 papers reviewed independently, then tagged by topic (e.g., "evaluation," "architecture," "HCI"). A single synthesis step aggregates by tag.

**Finding:** Flat aggregation detects fewer emergent patterns (6 vs. 16) because tags don't capture semantic relationships across papers. It also requires more review cycles (3.4 vs. 2.1) because cross-cutting issues aren't resolved at intermediate tiers, forcing re-reviews at the paper level.

#### 4.8.3 Ablation 3: Bidirectional vs. Unidirectional Flow

We test a unidirectional variant: issues bubble up ( $P1 \rightarrow PP1 \rightarrow B1$ ) but directives don't cascade down. Board generates B1 recommendations, but papers must manually interpret them.

**Finding:** Without downward propagation, B1 directive resolution drops from 89% to 54% in Round 1. Papers struggle to interpret abstract board recommendations ("improve methodological consistency") without concrete PP1/P1 tasks. Bidirectional flow ensures strategic decisions translate to actionable revisions.

### 4.9 Human Oversight Effectiveness

#### 4.9.1 Confidence-Based Deferral Results

Of 412 paper-tier priority items, 48 (12%) had confidence  $< 0.60$  and triggered human review. Human reviewers:

- **Confirmed** 32 (67%): AI classification was correct despite low confidence
- **Modified** 11 (23%): Changed priority level (e.g.,  $P1 \rightarrow P2$ )
- **Rejected** 5 (10%): Issue was spurious or out of scope

This prevented 5 false positives and 11 misclassifications from affecting revision plans. The correlation between low confidence and disagreement ( $r = 0.68$ ) validates the deferral mechanism.



## 4.9.2 Human Override Rates

Authors/leads overrode AI classifications in:

- **Paper tier:** 12/412 items (3%) — mostly P1 downgrades to P2 when authors had domain-specific knowledge
- **Panel tier:** 4/89 items (5%) — PP1 downgrades when panel chair judged pattern was local not systemic
- **Board tier:** 2/21 items (8%) — B1 rejections when board determined directive was premature

Post-override analysis: 85% of overrides were correct (human judgment superior to AI), 15% were debatable (borderline cases where either classification was reasonable).

## 5 Discussion

### 5.1 Why Hierarchical Structure Matters

The core finding is that **37-52% of high-priority issues are invisible at lower tiers**. This validates the fundamental premise: flat review structures miss emergent patterns that only appear when viewing multiple papers or modules together.

The escalation dynamics (Table 2) show that only 23% of P1 items escalate to PP1, and only 33% of PP1 items escalate to B1. This selectivity is crucial: the architecture amplifies signals that indicate systemic problems while filtering noise that represents local variation. If all P1 items escalated to PP1, the system would overwhelm higher tiers with irrelevant detail.

### 5.2 Bidirectional Flow vs. One-Way Escalation

Prior hierarchical systems (e.g., bug triage, incident management) primarily support **upward escalation**: local issues bubble up when they become severe. Our architecture adds **downward propagation**: strategic decisions at the board tier cascade as concrete tasks at the paper tier.

This bidirectional flow closes the loop from strategy to implementation. Without downward propagation, board directives remain abstract goals ("improve methodological consistency") without clear execution paths. With propagation, board directives generate specific PP1 tasks per module and P1 edits per paper, ensuring implementation.

The 89% resolution rate (Table 3) demonstrates effective propagation: most cascaded items are addressed within one review round. This suggests the system successfully translates high-level directives into actionable low-level tasks.

### 5.3 When Does the Architecture Provide Value?

The architecture provides the most value when:

- **Portfolio size:** 5+ papers per module, 2+ modules per program. Below this threshold, flat review suffices.
- **Cross-cutting concerns:** Papers share methodologies, datasets, or architectural components. If papers are fully independent, panel-tier synthesis provides little value.

- **Strategic coordination needed:** The research program has coherence requirements (shared vision, complementary contributions). Ad-hoc portfolios without strategic goals don't benefit from board-tier review.

For individual papers or small projects (1-3 papers), the paper tier alone is sufficient. The overhead of panel and board tiers (assembling reviewers, running cross-portfolio sessions) only pays off when cross-cutting patterns are likely.

## 5.4 Generalization to Other Domains

The architecture generalizes to any domain requiring multi-scale feedback:

**Code review:** Pull request reviews (paper tier), module owner reviews (panel tier), architecture committee reviews (board tier). Downward propagation ensures architectural decisions cascade to specific PRs.

**Grant evaluation:** Individual proposal reviews (paper tier), program officer panel discussion (panel tier), agency strategic review (board tier). Upward escalation surfaces themes (e.g., "5 proposals lack reproducibility plans") that inform program-wide policy.

**Product assessment:** Feature reviews (paper tier), product area reviews (panel tier), VP product review (board tier). Bidirectional flow aligns individual features with product strategy.

The key requirement is that entities at the lowest tier (papers, PRs, proposals, features) belong to meaningful groupings (modules, repositories, programs, product areas) that roll up to a strategic level (program, organization, company).

## 5.5 Limitations and Future Work

**Reviewer panel assembly:** We manually selected panel and board members for each tier. Automating this selection (matching expertise to module themes) is future work.

**Escalation threshold tuning:** The "3+ papers → PP1" rule is empirically derived. Different domains may require different thresholds.

**Real-time vs. batch operation:** The current system operates in batch mode (all papers complete paper tier, then panel tier activates). A real-time variant could trigger panel reviews incrementally as papers reach ready stage.

**Human-in-the-loop validation:** While the system generates PP1/B1 priorities automatically, a human review step could filter false positives (emergent patterns that aren't actually important).

**Cross-organizational deployment:** We evaluated the architecture within a single research program (14 papers, 3 modules). Scaling to multi-organization portfolios (e.g., conference PC meta-reviews) introduces new challenges: reviewer anonymity, conflicting priorities across organizations, lack of shared strategic vision.

## 5.6 Architectural Trade-offs

The three-tier architecture trades **complexity** for **pattern detection**. Adding panel and board tiers increases system complexity: more review rounds, more state tracking, more coordination overhead. This cost is justified when emergent patterns are common (37-52% of high-priority issues in our data). For domains where papers are truly independent, the cost exceeds the benefit.

The architecture also trades **latency** for **quality**. Papers must wait for all papers in the module to reach ready stage before panel tier activates, adding days or weeks to the review timeline. This serialization is necessary for cross-portfolio synthesis but delays individual paper feedback. A hybrid approach could provide incremental panel feedback as papers complete, trading synthesis completeness for lower latency.

## 5.7 Operational Considerations for Production Deployment

Deploying the architecture in production requires attention to observability, reliability, and cost [?].

### 5.7.1 Observability and Monitoring

Our deployment tracks 15 metrics (see Section 3.10) and implements anomaly detection:

- **Review quality drift:** One reviewer (Reviewer D) showed declining consistency over time ( $\sigma$  increased from 0.4 to 1.2 across rounds). Anomaly detection flagged this, prompting reviewer persona recalibration.
- **Synthesis quality:** Excessive P1 items ( $> 15$  per paper) triggered investigation, revealing over-sensitive semantic clustering (threshold lowered from 0.80 to 0.75).
- **Confidence trends:** Confidence scores declined 8% from Round 1 to Round 3, correlating with model API updates (GPT-4  $\rightarrow$  Claude 3.5). Dashboards surfaced this, prompting prompt recalibration.

Without observability, these issues would have degraded system quality silently. The 15-metric dashboard (Figure ?? in Appendix) provides real-time visibility into system health.

### 5.7.2 Testing and Regression Prevention

The three test suites (42 unit, 12 integration, 8 regression tests) caught 7 regressions during development:

- Prompt change broke P1/P2 boundary (classification shifted, caught by unit tests)
- Reviewer persona update changed scoring distribution (caught by golden dataset)
- State machine refactor introduced stage gate bug (caught by integration tests)

Test coverage (87-92%) ensures changes don't silently break functionality. Regression tests on golden datasets are critical: they provide ground truth for known-good reviews, catching drift when prompts/models change.

### 5.7.3 Cost and Performance

Per-paper costs (GPT-4 API, 2024 pricing):

- **Paper tier:** 5 reviews  $\times$  \$0.03/review + synthesis \$0.05 = **\$0.20/paper**
- **Panel tier:** 7 reviews  $\times$  \$0.04/review + synthesis \$0.08 = **\$0.36/module**
- **Board tier:** 7 reviews  $\times$  \$0.05/review + synthesis \$0.10 = **\$0.45/program**

For our 14-paper, 3-module evaluation: \$2.80 (paper) + \$1.08 (panel) + \$0.45 (board) = **\$4.33 total**.

Latency (wall-clock time):

- **Paper tier:** 2.3 minutes avg per paper (5 parallel review generations + synthesis)
- **Panel tier:** 5.1 minutes avg per module (7 parallel reviews + cross-paper synthesis)
- **Board tier:** 6.8 minutes avg (7 parallel reviews + cross-module synthesis)

Total wall-clock time for 14 papers: 32 minutes (paper tier) + 15 minutes (panel tier) + 7 minutes (board tier) = **54 minutes**. Parallelization across papers/modules reduces latency; sequential processing would take 4 hours.

### 5.7.4 Scalability Analysis

Our evaluation (14 papers, 3 modules) is **small-scale**. Projected costs at larger scale:

- **50 papers, 5 modules:** \$10 paper + \$1.80 panel + \$0.45 board = \$12.25 (\$0.25/paper)
- **100 papers, 10 modules:** \$20 paper + \$3.60 panel + \$0.45 board = \$24.05 (\$0.24/paper)

Cost scales linearly with papers (review generation), sublinearly with modules (panel tier amortizes across papers). Board tier cost is constant (single program).

**Human oversight scalability:** At 50 papers, if 12% of items trigger deferral, that’s  $50 \times 10 \text{ items/paper} \times 0.12 = 60$  human reviews needed. This is manageable for a research program. At 500 papers, 600 human reviews may exceed capacity, requiring higher confidence thresholds or hierarchical human review teams.

**Caveat:** Our evaluation does not validate the architecture at scales beyond 20 papers per module or 5 modules per program. Claims about “scaling” should be interpreted as scaling *from single-paper review to multi-paper portfolios*, not scaling to arbitrarily large systems.

## 5.8 Limitations and Threats to Validity

**External validity:** Our evaluation is on a single research program (14 papers, AI/systems domains). Generalization to other domains (biology, social science, humanities) requires validation. The architecture may work differently for papers with different structure (empirical vs. theoretical) or review culture (anonymous vs. open).

**AI reviewer calibration:** Our reviewer personas are based on public work of named researchers. Persona accuracy depends on prompt quality and available training data. Reviewers from underrepresented communities or emerging fields may be harder to simulate accurately.

**Emergent pattern validation:** While we validated emergent patterns against human panels (75% precision, 82% recall), our validation set is small (8 papers). Larger validation studies are needed to establish generalizability.

**Longitudinal effects:** We evaluated the system over 4 months (Jan-Apr 2026). Longer-term effects (reviewer drift, author learning, system gaming) are unknown. Continuous monitoring is required to detect degradation.

**Comparison to human-only review:** Our baseline is flat AI review, not human expert review. We show the three-tier AI architecture outperforms flat AI, but do not claim it matches or exceeds human expert panels. The 73% agreement rate (Section 4.7) suggests the system is complementary to, not a replacement for, human review.

## 6 Conclusion

We presented a three-tier hierarchical review architecture for scaling AI-simulated expert feedback from individual papers to multi-module research programs. The architecture addresses a fundamental limitation of flat review systems: cross-cutting patterns and strategic misalignments are invisible at the paper level.

Evaluation on 14 papers across 3 modules demonstrates that 37% of module-level and 52% of program-level high-priority issues are emergent patterns not flagged by any individual paper review. The bidirectional flow protocol successfully escalates issues upward (23% of P1 items  $\rightarrow$  PP1) and propagates directives downward (89% resolution rate in one round), closing the loop from strategy to implementation.

The architecture reduces paper-tier review cycles by 31% by resolving cross-cutting issues once at higher tiers rather than redundantly at each paper. It generalizes to any domain requiring multi-scale structured feedback: code review, grant evaluation, product assessment.

Key contributions:

1. First hierarchical architecture for AI review systems with formal priority escalation rules (P1/P2/P3  $\rightarrow$  PP1/PP2/PP3  $\rightarrow$  B1/B2/B3).
2. Empirical evidence that emergent patterns constitute 37-52% of high-priority issues at higher tiers.
3. Reusable implementation pattern with bidirectional flow protocols applicable to multiple domains.

Future work includes automating reviewer panel assembly, tuning escalation thresholds per domain, and extending the architecture to real-time incremental review. The system is open source and available as a Claude Code plugin.

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**AI Simulation Disclosure:** This research involves AI-simulated expert review personas based on published work of real researchers. These are not actual reviews from the named individuals. See <https://github.com/panel-review/methodology> for full methodology disclosure.

## References